AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

LECTURE 3

BIOETHANOL

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COURSE STUDY GUIDE

Date	Meeting/ Week #	Topic for the day	Activity
14-16 Jan	1		Lecture
19-23 Jan	2	Introduction to Energy Systems	Lecture
26-30 Jan		Renewable Energy Systems and their Potential	
2-6 Feb	3	Bioethanol	Lecture
9-13 Feb	4	Hydropower / Group assignments	Lecture
16-20 Feb	5	Biodiesel	Lecture
23-27 Feb	6	Biogas	Lecture
2-6 Mar	7		Lecture
9-13 Mar	8	Mid-semester exams	Quiz
16-20 Mar	9	Discuss group assignment challenges	Lecture
23-27 Mar	10	Solar Energy	Lecture
30 – 3 Apr	11	Wind Energy	Lecture
6-10 Apr	12	Renewable Energy Applications in Agriculture	Lecture
13-17 Apr	13	GROUP PRESENTATIONS	Student-led
13-17 April	14	GROUP PRESENTATIONS	Student-led

Housekeeping

- I am away next week.
- Hydropower lecture will take place on the weekend, Saturday to be precise.
- Date is 14 February and time is 10-12 noon

Lecture outline

- Bioenergy
- Biofuels
- **BIOETHANOL**
  - Bioethanol production
  - Ethanol from sugar feedstock
  - Ethanol from starch feedstock
  - Step by step ethanol production process
  - Uses of Bioethanol
  - Economics of bioethanol

Bioenergy

- Bioenergy refers to the conversion and use of plant and other organic materials (biomass) to provide desired forms of energy and energy services such as electricity, liquid or gaseous fuels and process heat (Kartha *et al*, 2005).
- Since the discovery of fire, biomass has been a very important energy source for mankind. Besides its direct use as firewood and charcoal in developing countries, the use of biomass in the form of biofuels is the next important application of bioenergy.
- The main application for biofuels is as fuel for transportation, however, they are also used in stationary diesel and gasoline engines

Biofuels

- Biofuel is any fuel that is derived from biomass, living organisms or their metabolic by-products, such as manure from cows.
- Biofuel is produced by processing biological materials to generate fuels such as biodiesel and bioethanol, and biogas (using anaerobic digestion to generate methane from biodegradable materials and wastes)
- Biofuels can be used to produce an array of energy-related products, including electricity, liquid, solid, and gaseous fuels, heat, chemicals, and other materials.

What is bioethanol?

- Ethanol (CH_3CH_2OH) belongs to the group of chemical compounds whose molecules contain a hydroxyl group, $-OH$, bonded to a carbon atom.
- It is a clear, colourless liquid with a characteristic, agreeable odour.
- The terms ‘fuel ethanol’ and „ethanol fuel‟ are usually used in literature to refer to ethanol used as a transportation fuel.
- ‘Bioethanol’ is ethanol that is obtained from the conversion of carbon-based feedstocks that are considered renewable.

Bioethanol feedstock

- Sugarcane
- Sugar beet
- Sweet sorghum
- Corn
- Cassava
- Wheat
- Cellulose











Bioethanol feedstock

- Sugarcane and sweet sorghum are voluminous crops and their transportation is costly.
- Meanwhile, they must be processed within forty-eight hours after harvesting, otherwise, energy yields start to decrease significantly. For these reasons, these feedstocks should be produced, as much as possible, in the vicinity of the ethanol plant.

Bioethanol feedstock

- As a biofuel crop, cassava has the advantage that the root can remain in the ground for months without deterioration, allowing for carefully planned and continuous harvest schemes (Sielhorst, *et al*, 2008).
- Once it is harvested, cassava needs to be processed within two to three days and makes it possible for cassava to be grown at considerable distance from the ethanol plant. The roots are not as voluminous as sugarcane or sorghum stalks and cassava is therefore easier and cheaper to transport (Sielhorst, *et al*, 2008).

Bioethanol production

- Ethanol could be produced through chemical reactions or by conversion of biomass materials by fermentation processes following three major steps:
  - the formation of a solution of fermentable sugars;
  - the fermentation of these sugars to ethanol;
  - and the separation and purification of the ethanol, usually by distillation

Production processes

- The production processes depend on whether the feedstock is a sugar crop (sugar cane and sweet sorghum), or starch crop (grain and cassava).
- The principal differences have to do with the initial processes the feedstock goes through before the fermentation stage.
- But no matter the feedstock used, the final stage of ethanol production is fermentation.
- Sugar crops follow the same process line whereas starch crops can be converted to ethanol using the dry-milling or wet-milling production route.

Ethanol from sugar feedstock

- Raw cane (from the sugarcane or sweet sorghum plant) is washed, chopped, and crushed in rolling mills to separate the sugar-laden juice (sucrose) from the fibre in the cane, called bagasse.
- The juice, which contains over 90 percent of the sucrose in the cane, is filtered to remove impurities. Filtered juice is heated and fermented and then fed to a distillery.

Fermentation

- In the fermentation stage, yeasts and other microbes are added to produce a dilute alcohol product with ethanol concentration of 10 – 15%.
- The fermented solution is then distilled to remove most of the water from the mixture to give “hydrous ethanol” with about 95% ethanol content and 5% water.
- A final dehydration step is required to remove the remaining water to produce “anhydrous ethanol” that can be blended with gasoline.

Ethanol from starch feedstock

Ethanol from starch crops is produced by one of two processes: **wet milling** and **dry milling**.

Dry milling

- The dry milling process involves cleaning and breaking down the cereal kernel into fine particles using a hammer milling process. This creates a powder with a coarse flour type consistency.
- In order to produce a sugar solution the mixture is hydrolysed or broken down into sucrose sugars using enzymes or a dilute acid.
- The mixture is then cooled and yeast is added in order to ferment the mixture into ethanol.
- The dry milling process is normally used in factories producing less than 50 million gallons of ethanol every year

Wet milling

- In the wet milling process, corn kernel is steeped in a solution of water and sulphur dioxide (SO2); this helps to break down the proteins and release the starch present in the corn and helps to soften the kernel for the milling process.
- The corn is then milled to produce germ, fibre and starch products. The germ is extracted to produce corn oil and the starch fraction undergoes centrifugation and saccharification to produce gluten wet cake.
- The ethanol is then extracted by the distillation process. The wet milling process is normally used in factories producing several hundred million gallons of ethanol every year

Principal differences between wet and dry mill ethanol plants

Parameter	Dry milling	Wet milling
Cost of construction	Lower	Higher
Versatility of products	Less versatile	More versatile
Co-products	Dried distillers grain and corn meal	Sweeteners, corn oil, gluten feed and gluten meal
Value of co-products	Less valuable	More valuable
Yield per feedstock	Higher	Slightly Less

Step by step ethanol production process

- **Milling:** The feedstock (corn) is passed through a hammer mill and is grinded into a fine meal to free the starchy material from the grain kernel and to allow water access to all the starch in the grains.
  - It is much like the grinding process that agricultural residues go through so that the particles have uniform size and to make the production process more efficient.

Step by step ethanol production process

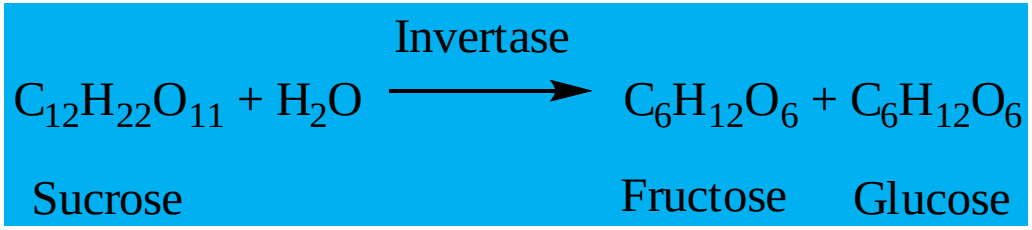
- **Liquefaction:** The meal is mixed with warm water to form a mash. An enzyme (alpha-amylase) is added and heat is applied to enable liquefaction. The mixture is cooked at a high temperature of between 120 – 150 ° C. The temperature is then lowered to about 95° C. The initial higher temperature enables the elimination of possible bacteria and the lower holding temperature ensures that the mixture is still in the liquid state.

Step by step ethanol production process

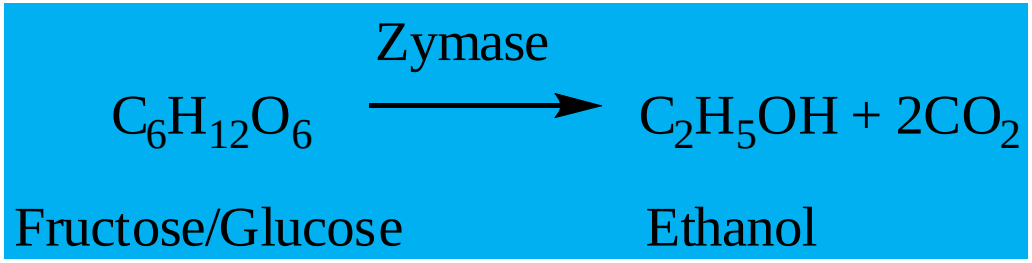
- **Saccharification:** The mash from the preceding process is cooled to 60 ° C and a secondary enzyme (gluco-amylase) is added. The purpose of this enzyme is to convert the starch to fermentable sugars, also known as dextrose. The temperature keeps the reaction progressing fast enough while at the same time avoiding unwanted side reactions. Saccharification in a wet mill may take up to 48 hours, though it usually requires less time, depending on the amount of enzyme used.

Step by step ethanol production process

- **Fermentation:** The mash is then further cooled to a fermentation temperature of about 32 ° C. Yeast is added to ferment the sugars to ethanol and carbon dioxide (CO2) and small quantities of other organic compounds. The yeast contains an enzyme called invertase, which acts as a catalyst and helps to convert the sucrose sugars into glucose and fructose (both C6H12O6).



- The fructose and glucose sugars then react with another enzyme called zymase, which is also contained in the yeast to produce ethanol and carbon dioxide.



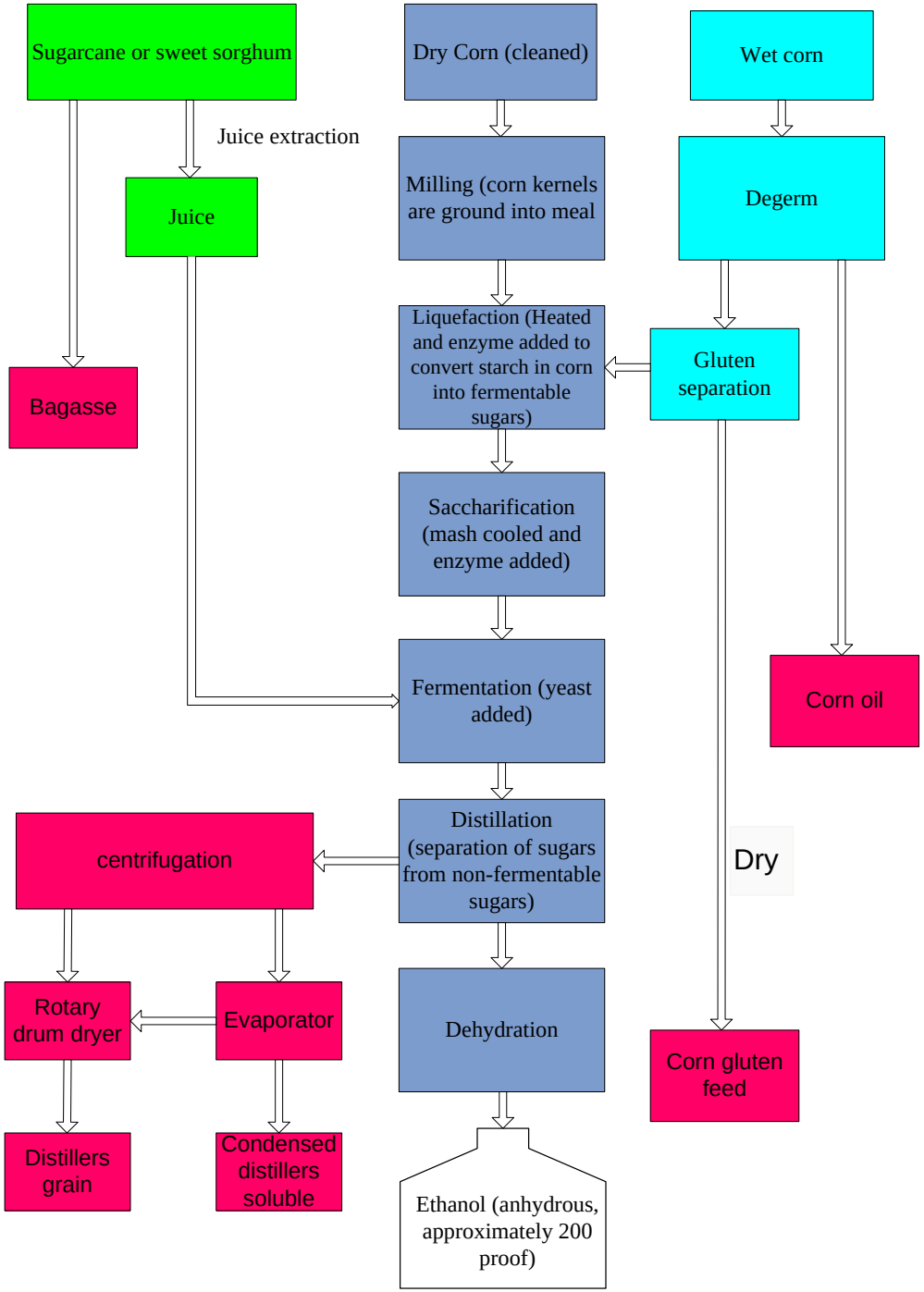
Step by step ethanol production process

- **Distillation**: The fermented mash (beer) contains significant quantity of water, as well as all the non fermentable solids from the feedstock and the yeast cells which must be removed. This is achieved through distillation. The distillation process works by boiling the water and ethanol mixture. Since ethanol has a lower boiling point (78.3 ° C) compared to that of water (100 ° C), the ethanol turns into the vapour state before the water and can be condensed and separated. The ethanol leaves the top of the final column at about 95 – 96% strength (ethanol and water form a 95 – 96% azeotrope so it is not possible to reach 100% by simple distillation).

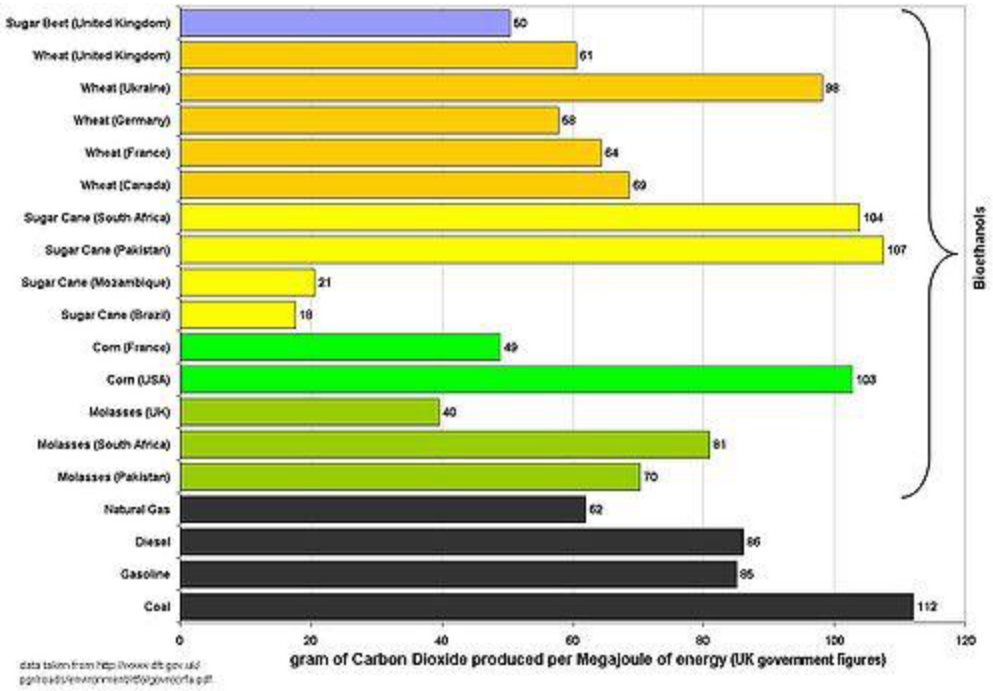


Step by step ethanol production process

- **Dehydration:** The ethanol from the top of the column is passed through an absorber containing a molecular sieve (dehydration system) to remove the remaining water. The ethanol at this stage is called anhydrous (pure, without water) ethanol and is approximately 200 proof. Three absorbers may be used to enhance efficiency. A drink that‟s 100 proof is 50% alcohol, while a 200 proof drink is 100% pure alcohol.
- **Denaturing:** Fuel ethanol is at this stage denatured with a small amount (2 – 5%) of some product such as gasoline, to make it unfit for human consumption.


Sugarcane or sweet sorghum
Juice extraction
Juice
Bagasse
Dry Corn (cleaned)
Milling (corn kernels are ground into meal
Liquefaction (Heated and enzyme added to convert starch in corn into fermentable sugars)
Saccharification (mash cooled and enzyme added)
Fermentation (yeast added)
Distillation (separation of sugars from non-fermentable sugars)
Dehydration
Ethanol (anhydrous, approximately 200 proof)
Wet corn
Degerm
Gluten separation
Corn oil
Dry
Corn gluten feed
centrifugation
Rotary drum dryer
Evaporator
Distillers grain
Condensed distillers soluble

Properties of Bioethanol



Properties of bioethanol

Property	Comment
Solubility in water	Fuel ethanol will mix with water, but at high enough concentrations of water, the ethanol will separate from the water
Energy content	For identical volumes, ethanol contains less energy than gasoline. On an energy basis, 1.0 gallon of E85 is equivalent to 0.72 gallons of gasoline
Flame visibility	A fuel ethanol flame is less bright than a gasoline flame but is easily visible in daylight
Specific gravity	Pure ethanol and ethanol blends are heavier than gasoline
Conductivity	Ethanol and ethanol blends conduct electricity. Gasoline, by contrast, is an electrical insulator
Toxicity	Ethanol is less toxic than gasoline or methanol. Carcinogenic compounds are not present in pure ethanol; however, because gasoline is used in the blend, E85 is considered to be potentially carcinogenic
Flammability	At low temperatures, the vapour concentration in an E85 tank can fall into the flammable range The temperature for flammable E85 vapours depends on the E85 volatility. Although less likely than with E85, gasoline tanks can also contain flammable vapours at extremely low temperature

Bioethanol Application technologies



Spark ignition engines

- The principal application of bioethanol up to date has been in the spark ignition engine as a transportation fuel.
- The energy content is however lower than that of petrol (for pure ethanol, about two-third that of petrol)
- Bioethanol is also corrosive when used as a pure fuel, but low blends with petrol helps avoid this problem.
- Bioethanol can be used in blends of between 10 and 20% with fossil gasoline without necessary engine modifications. In Brazil, all brands of automotive gasoline contain anhydrous ethanol in the range of 20−25 %

Biofuels sustainability challenges

1. The large-scale production of bioenergy feedstocks pose a threat to ecosystems, soil quality and water use.
   - Demand for bioenergy could increase the pressure for deforestation by requiring more land for energy crops.
   - This can contribute to soil erosion, increase drought risks, and affect local biodiversity.





Biofuels sustainability challenges

2. The large-scale production of bioenergy may divert agricultural production away from food crops and drive prices up.
   - Energy crops, if grown on a large scale, may compete with food crops in a number of ways including land use, investment requirements, infrastructure support, water, fertilizers, etc.





Sustainability challenges

3. The influx of large multinational companies into the bioenergy industry may simply replace unsustainable bioenergy production, maximising profit but yielding undesirable social and environmental outcomes.

4. A dramatic increase in the production and use of bioenergy has the potential to significantly reduce overall GHG emissions associated with energy consumption; alternatively, it could intensify the threat of global warming.

Decision tree for strategic national choices on biofuel development

